

DALBANDIN, Pakistan

WMO: 417120

Lat: 28.883N

Lon: 64.400E

Elev: 2789

StdP: 13.27

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.3	33.7	9.0	9.7	56.6	12.4	11.4	50.7	14.9	66.5	11.4	60.7	1.5	50	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.0	112.9	71.2	111.0	70.6	109.2	69.9	77.7	105.1	75.0	103.3	73.0	102.8	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.3	119.5	98.1	65.5	104.5	97.4	61.9	91.8	96.6	43.4	105.0	40.5	103.5	38.6	102.9	93.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.8	9.1	7.5	25.1	116.6	4.9	1.8	21.6	117.9	18.7	118.9	15.9	120.0	12.4	121.3	
(5)				20.8	80.4	3.2	5.9	18.5	84.6	16.6	88.1	14.8	91.4	12.5	95.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	75.9	52.3	57.7	68.1	79.3	89.0	95.6	97.3	93.0	86.0	75.7	62.6	52.8	(6)
(7)		DBStd	17.06	6.77	8.83	7.27	6.14	4.90	3.32	3.01	3.29	4.51	5.91	7.03	6.39	(7)
(8)		HDD50	134	55	32	0	0	0	0	0	0	0	0	3	44	(8)
(9)		HDD65	1183	393	228	55	2	0	0	0	0	0	1	126	378	(9)
(10)		CDD50	9580	127	246	561	878	1209	1367	1466	1334	1081	797	382	132	(10)
(11)		CDD65	5156	1	23	151	430	744	917	1001	869	631	333	55	1	(11)
(12)		CDH74	83154	48	324	1884	5688	11735	16097	17872	14672	9497	4420	852	65	(12)
(13)		CDH80	56162	2	72	745	3210	7800	11810	13420	10370	6148	2311	267	7	(13)
(14)	Wind	WSAvg	2.9	2.6	3.0	3.2	3.0	3.2	3.5	3.2	2.9	2.5	2.4	2.4	2.5	(14)
(15)	Precipitation	PrecAvg	3.3	0.6	0.6	0.7	0.3	0.1	0.0	0.2	0.0	0.0	0.1	0.1	0.4	(15)
(16)		PrecMax	8.5	2.8	2.4	3.3	1.9	0.6	0.4	3.4	0.7	0.1	0.9	1.7	3.2	(16)
(17)		PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.9	0.7	0.7	0.8	0.4	0.1	0.1	0.5	0.1	0.0	0.2	0.2	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.1	86.3	95.2	103.9	111.6	115.2	115.5	112.8	106.9	102.4	90.3	80.5	(19)
MCWB			55.1	59.2	62.2	64.9	70.3	70.5	71.5	73.7	67.1	66.6	59.9	55.6	(20)	
2%		DB	74.8	81.9	90.9	99.9	108.7	112.8	113.0	109.9	104.9	98.5	86.6	75.5	(21)	
		MCWB	54.3	57.3	61.4	65.1	69.0	70.7	71.3	70.3	66.5	63.2	60.2	54.8	(22)	
5%		DB	71.3	78.1	87.4	97.2	106.0	110.7	111.3	107.9	103.1	94.9	83.1	71.4	(23)	
		MCWB	52.3	55.9	59.8	64.9	67.8	70.0	70.8	69.3	65.9	62.2	58.5	52.3	(24)	
10%		DB	67.5	74.2	84.3	94.2	103.3	108.5	109.5	105.9	101.0	91.3	79.0	67.5	(25)	
		MCWB	50.2	54.3	58.4	63.5	66.5	69.2	70.7	68.1	64.9	61.0	56.5	50.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.7	60.7	66.2	71.0	78.3	78.9	82.5	80.5	73.0	69.8	65.4	58.4	(27)
MCDWB			73.9	82.6	89.3	97.3	107.3	108.2	108.7	105.9	99.1	95.9	86.1	73.7	(28)	
2%		WB	55.5	58.5	63.3	68.5	72.2	75.2	78.5	75.8	70.5	66.8	61.9	55.5	(29)	
		MCDWB	71.4	78.7	85.5	93.2	100.8	106.0	104.0	103.1	99.8	91.3	81.7	72.9	(30)	
5%		WB	53.7	56.9	61.6	66.4	69.6	72.6	75.5	73.5	68.3	64.6	59.6	53.5	(31)	
		MCDWB	69.1	76.0	83.3	92.9	101.3	105.4	103.2	99.7	99.8	90.5	79.4	68.4	(32)	
10%		WB	51.7	55.1	59.9	64.6	67.5	70.7	73.4	71.4	66.1	62.5	57.5	51.5	(33)	
		MCDWB	65.1	72.2	80.7	91.1	99.7	105.0	104.2	99.6	98.3	89.0	77.3	65.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	26.6	25.8	26.5	26.5	25.8	26.2	25.0	26.3	29.7	30.3	29.6	28.3	(35)	
5% DB		MCDBR	31.9	29.6	30.0	29.5	28.9	28.4	26.8	28.2	30.9	31.8	33.3	32.8	(36)	
		MCWBR	17.3	14.0	12.6	12.1	11.7	10.9	10.4	11.0	12.8	13.4	17.1	17.5	(37)	
5% WB		MCDBR	27.5	28.2	28.2	27.8	27.3	27.4	24.9	24.3	29.9	30.4	30.9	29.6	(38)	
		MCWBR	15.6	14.2	13.8	13.2	12.8	12.2	11.8	11.3	14.3	16.0	17.4	16.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.351	0.410	0.457	0.517	0.532	0.530	0.604	0.541	0.434	0.398	0.359	0.334	(40)	
taud		2.272	2.086	1.941	1.830	1.797	1.803	1.680	1.790	2.001	2.095	2.241	2.321	(41)		
Ebn at Noon		278	269	263	252	247	246	228	242	266	268	272	279	(42)		
Edh at Noon		36	47	58	67	70	69	78	69	54	46	37	33	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1244	1466	1851	2148	2356	2473	2302	2246	2097	1767	1406	1204	(44)	
(45)		RadStd	77	101	78	97	70	100	111	84	53	39	44	59	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page